PROCESS CELL: PC_CHI_PACK — Packaging Process Cell (CHI — Chicago | NA Site)

SITE HIERARCHY — PC_CHI_PACK

PC_CHI_PACK → UNIT_CHI_PACK → EM_CHI_PACK_MAIN + EM_CHI_PACK_QA

FLOW WRAPPER ZONE

PALLETIZER ZONE

TT 001 — WRAP_SEAL_TEMP_DEGC

PT 002 — FLOW_WRAPPER_PRESSURE_

ST 003 — WRAP_SPEED_MPM

IT 004 — WRAP_MOTOR_CURRENT_A

ST 011 — PALL_SPEED_RPM

CT 012 — PALL_COUNT_PALLETS

WI 013 — PALL_WEIGHT_KG

IT 014 — PALL_MOTOR_CURRENT_A

FROM PC_CHI_COOL

(COOLED COOKIES)

FILM REEL

LDPE FILM

EQ_CHI_018

CHI Flow Wrapper

(Flow Wrapper)

SEAL JAWS

Mfr: Rockwell Automation

OS 018 — WRAP_OPERATING_STATE

AS 019 — WRAP_ALARM_STATE

OUTFEED / ACCUMULATION CONVEYOR

EQ_CHI_019

CHI Palletizer

(Palletizer)

PALLET

Mfr: Rockwell Automation

OS 018 — PALL_OPERATING_STATE

AS 019 — PALL_ALARM_STATE

TO WAREHOUSE / SHIPPING

(PACKAGED PALLETS)

REJECT/WASTE

WT 005 — WRAP_WEB_TENSION_N

CT 006 — WRAP_COUNT_RATE_PCS_MI

FT 007 — WRAP_FILM_FEED_M_MIN

LT 015 — PALL_LAYER_COUNT

RT 016 — PALL_REJECT_RATE_PCS_M

PLC CABINET — UNIT_CHI_PACK / EM_CHI_PACK_MAIN

CM_COOKIE_004 — Flow Wrapper — Seal 1 — *[MotorCtrl]* — PLC-04/DB1203

CM_COOKIE_012 — Flow Wrapper — Seal 2 — *[BatchCtrl]* — PLC-12/DB1211

CM_COOKIE_020 — Flow Wrapper — Seal 3 — *[SEQ]* — PLC-04/DB1219

CM_COOKIE_028 — Flow Wrapper — Seal 4 — *[MotorCtrl]* — PLC-12/DB1227

EM_CHI_PACK_QA — Quality Assurance / Safety Module

EQ_AUX_006 — IFM

EQ_AUX_020 — Danfoss

EQ_AUX_034 — Eaton

EQ_AUX_048 — Omron

VFDs: EQ_AUX_005 · EQ_AUX_019 · EQ_AUX_033 · EQ_AUX_047

NOTES:

1. Process cell PC_CHI_PACK (CHI — Chicago | NA Site): FROM PC_CHI_COOL → Flow Wrapper → Palletizer → Warehouse.

2. Flow Wrapper EQ_CHI_018 (Rockwell Automation) wraps cookies in LDPE film; sealing controlled by PLC.

3. Palletizer EQ_CHI_019 (Rockwell Automation) builds and wraps pallet loads.

4. VFDs (EQ_AUX_005, EQ_AUX_019, EQ_AUX_033, EQ_AUX_047) provide variable speed for wrappers and conveyors.

5. TS_-_PACK_FLOW_WRAPPER_PRESSURE_BAR is the only explicitly modelled CDF time-series for this cell.

6. State encoding: 0=STOPPED 1=RUNNING 2=IDLE 3=STARTUP 4=ALARM 5=MAINTENANCE.

LEGEND

- TT — Seal Bar Temperature
- PT — Seal Jaw Pressure
- ST — Speed Transmitter
- IT — Motor Current Indicator
- WT — Web Tension
- CT — Count Rate
- FT — Film Feed Rate
- WI — Pallet Weight
- LT — Layer Count
- RT — Reject Rate
- OS — Operating State
- AS — Alarm State